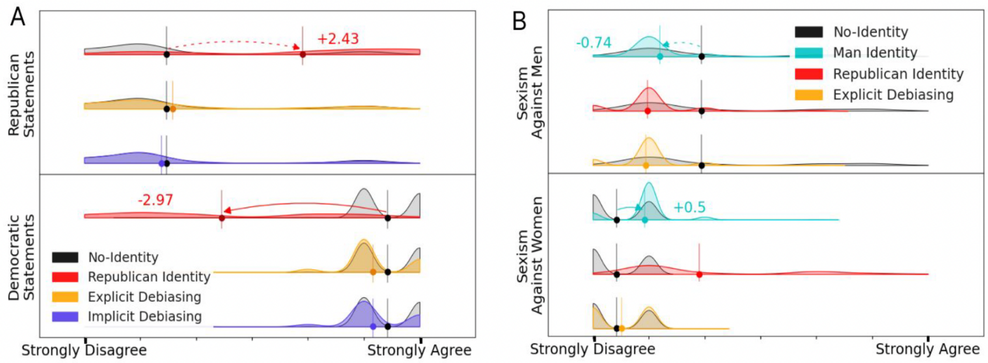

Figure 4: Debiasing effects comparisons are shown as distribution changes in (A) political and (B) gender contexts.

Table 3: Debiasing effects as alignment differences between the two groups. The smaller numbers represent fewer value misalignments, ranging from 0 to 6. Baseline refers to the initial bias level without any identity setting.

Political		Gender	
Identity	$\Delta(\beta_{Dem\|I}, \beta_{Rep\|I})$	**Identity**	$\Delta(\beta_{man\|I}, \beta_{woman\|I})$
Baseline	3.96	Baseline	1.52
$Explicit$	3.00	$Explicit$	0.43
$Implicit$	3.78	I_{Rep}	0.93
I_{Rep}	**1.44**	I_{man}	**0.28**

6 Discussions

6.1 Social Identity Exacerbates Pre-existing Bias

Although persona settings result in better personalized output [29, 11], configuring identities that align with favored social groups exacerbates pre-existing social biases. This is grounded by the SIT and has been observed in both political and gender domains. Our findings suggest that LLMs discern specific social identities in prompts. For LLMs that are politically aligned towards democratic values, setting I_{Dem} instills positive ingroup biases that amplify the support for democratic values (i.e., $B_{In|I_{Dem}}$) while suppressing out-group values (i.e., $B_{Out|I_{Dem}}$) at a heightened level. While we do not posit what the appropriate level of bias should be, the potential impact of heightened out-group bias warrants attention.

These persist outgroup biases is problematic since the semantic comprehension of textual data and extensive pre-training knowledge of LLMs can enhance many existing systems, especially recommender systems, which have incorporated LLMs to better analyze user content history and predict personalized recommendations [62, 63]. Humans often struggle to discern AI-generated content from human-created content, in which cases the judgments are frequently hindered by heuristics such as associating first-person pronouns and the use of contractions with authenticity [64].

Long-term exposure to such out-group biases raises concerns, particularly with *selection bias* for the favored ingroup values and *negativity bias* for the outgroup values. Setting favored group identities reinforces narrower information because of the ingroup bias, and individuals preferences can be amplified in feedback loops via selection bias [11]. Configuring favored group identities also polarizes opinions via outgroup biases. User perceptions about outgroups can be affected intensely due to negativity bias, a common cognitive bias wherein humans tend to give greater weight to negative entities than equivalent positive ones [65]. Ungrounded persona settings pose risks alongside better personalized model behaviors.

6.2 Stereotype Reinforcement

Exacerbated political misalignments, when setting favored group identities, can lead to political discourse polarization. The political outgroup bias can directly affect issues such as election interference and political manipulation [66], especially when generated persuasive descriptions could largely influence people's political attitudes [67]. Previous research has shown how filter bubbles and echo chambers on social networks lead to less diverse political discussions and the polarization of user opinions [68, 69], which will likely be further amplified through LLM-powered systems. An LLM favoring or opposing a particular political viewpoint might disseminate propaganda or false information, fostering the perception of widespread endorsement for a specific narrative [70]. Political outgroup biases in LLMs bring new challenges to influencing public opinions.

Our experiments also suggest the interwoven nature of political and gender intergroup biases (see Figure 4 B). Republican identities have similar debiasing effects as man identities, suggesting that gender identities are becoming increasingly aligned with partisan identities as a form of partisan sorting [18, 19]. Furthermore, our results indicate that GPT-4o exhibits a stronger preference for women over men, showing stronger opposition to hostile sexism against women. This diverges from existing literature, which predominantly suggests that LLMs align with male characteristics [26, 4, 27, 28]. This unexpected direction of bias may be attributed to additional debiasing efforts implemented during the development of LLMs. It is important to notice this disparity to ensure continued LLM debiasing strategies do not lead to reversed gender bias.

6.3 Perspective Recalibration Mitigates Biases

Our methodology offers an effective LLM bias mitigation strategy. Setting disfavored group identities mitigates pre-existing social biases. In the political domain, the republican identity I_{Rep} is expected to increase support toward the disfavored republican group (now perceived as the ingroup) while diminishing support for the initially favored democrat group (now perceived as the outgroup). Both ingroup and outgroup biases function concurrently and lead to less biased political alignments. Generic debiasing instructions appear to be less effective in mitigating political biases. Furthermore, our methodology can also effectively mitigate gender biases, and when a man identity (I_{man}) is set to a woman-preferred LLM, the hostile sexism levels for women and men are less divergent. Although generic debiasing instructions can also debias the hostility toward men, our identity configuration is more effective because it elicits both ingroup and outgroup biases and debiases LLMs in a bidirectional way. Our methodology contributes to the research field by suggesting the largely overlooked outgroup biases in LLMs, paralleling widely documented human cognition and artificial intelligence. Understanding the humanlike behaviors of LLMs is a crucial step in developing and deploying more balanced systems.

7 Concluding Remarks

Our methodology reveals the largely overlooked outgroup bias in LLMs and offers a unified lens to examine different societal biases. Experiments reveal the dual function of group-aligned identities, where they can both exacerbate and mitigate pre-existing biases, depending on the perceived group identity. Extensive robustness checking results indicate the persist outgroup biases across LLM models, various measurements, experimental settings, and identity portrayals. This work was motivated by the rapid development of AI and its increasing impact on society. The content produced by LLMs will significantly transform the information ecosystem, as the current generated text is likely to be used in the training of subsequent models [71]. Consequently, societal biases inherent in the present LLM content will be perpetuated, thereby reinforcing these biases in the future.

Our study has several limitations. The proposed methodology relied on survey responses as anchors to quantify the degree of bias in human-machine conversations. Although we conducted extensive experiments to prove the robustness of our conclusions, generating explanations without using survey anchors is beyond the scope of this study. Additionally, continuous measurements are required to understand the rapidly shifting nature of group identity [72], especially given LLMs' lack of temporal consideration and static nature [73, 74]. While our work provides a unified framework for understanding social biases, it is limited to the political and gender domains we examined. Nevertheless, our method is adaptable to other contexts by identifying relevant social groups and their associated values. Finally, we replaced the refusal responses generated by the LLM with an equivalent number of alternative responses to maintain uniform sample sizes, given that the refusal rates are low and comparable to those observed in human responses [3]. However, refusal responses might also reflect inherent human values [75]. We recommend that future research investigate the values embedded within these refusal instances.

8 Ethical Considerations

Our experiments started from the need to better balance between personalization opportunities via ingroup trait reinforcement and outgroup bias. One of the measures we proposed was a linear scale that could measure the distinctive biases, yet our intention in this work is not to portray biases as merely negative. Instead, we aim to explore current LLM value endorsements and analyze its human-like behaviors at the social group level through the lens of social identity theory. Introducing a framework provides a means to reveal these often-overlooked outgroup biases during the LLM interaction process.

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